

Lesson 4: The Limits of Algorithms

45 minutes

Overview

Students explore the limits of what algorithms are able to compute. First they use a widget that exposes students to the Traveling Salesperson Problem. After recognizing this problem can only be solved using an unreasonable time algorithm the develop their own good enough solutions that run more efficiently. Later in the lesson students watch a video about undecidable problems for which no algorithm can ever be developed to solve them.

Standards

Full Course Alignment

CSP Conceptual Framework

- ▶ **AAP-4** - There exist problems that the computer cannot solve, and even when the computer can solve a problem, it may not be able to do so in a reasonable amount of time.

CSTA K-12 Computer Science Standards (2017)

- ▶ **AP** - Algorithms & Programming

Agenda

Warm Up (5 minutes)

Activity (30 minutes)

Traveling Salesperson

Undecidable Problems

Wrap Up (10 minutes)

Assessment: Check For Understanding

Objectives

Students will be able to:

- Determine if an algorithm runs in unreasonable time.
- Develop a heuristic to solve a problem.
- Distinguish between decision problems and optimization problems.
- Explain the existence of undecidable problems

Preparation

- Read through the slides
- Preview the Traveling Salesperson widgets
- Check the "**Teacher's Lounge**" forum for verified teachers to find additional strategies or resources shared by fellow teachers
- If you are teaching virtually, consider checking our **Virtual Lesson Modifications**

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **The Limits of Algorithms** - Slides
- ▼ Make a Copy

For the students

- **The Halting Problem: The Unsolvable Problem** - Video

Vocabulary

- **Decision Problem** - a problem with a yes/no answer (e.g., is there a path from A to B?)
- **Heuristic** - provides a "good enough" solution to a problem when an actual solution is impractical or impossible
- **Optimization Problem** - a problem with the goal of finding the "best" solution among many (e.g., what is the shortest path from A to B?)
- **Undecidable Problem** - a problem for which no algorithm can be constructed that is always capable of providing a correct yes-or-no answer

Teaching Guide

Warm Up (5 minutes)

 **Discuss:** *What is the difference between a reasonable and unreasonable time algorithm?*

Discussion Goal: Use this discussion to quickly review topics brought up in the previous lesson.

- Reasonable algorithms grow at a polynomial rate or slower. Unreasonable algorithms grow exponentially.
- The time to solve an unreasonable algorithm grows very quickly even for relatively small problem sizes.

Have students think about their answers on their own, then share with a partner, and then finally discuss responses with the entire room.

Remarks

This was a good reminder of what we learned last time. Unreasonable time algorithms just don't make sense to run. Today we're going to be thinking more about what happens when we reach the limits of algorithms, and how sometimes we can make compromises.

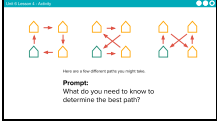
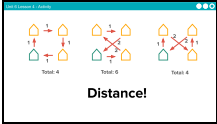
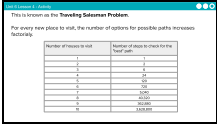
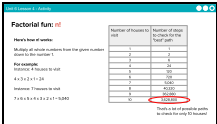
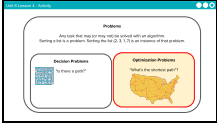
Activity (30 minutes)

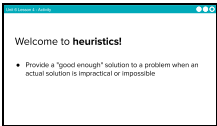
Traveling Salesperson

 **Set Up:** Each student needs to have their journals and a pen or pencil.

 **Display:** Use the activity slides for this lesson to guide the activity on the limits of algorithms.

Slides	Speaker Notes
	Say: Today we are going to explore the Traveling Salesperson Problem.

Slides	Speaker Notes
	Discuss: How many different paths can you find to visit all of your friends' houses? Do This: Students should sketch out different paths in their journals. Read through the rules together as a class. Rules: - You must start and end at your own house. - You can only visit each house once. Discuss: Students share their paths with a partner before discussing as a class.
	Say: Take a look at some of these paths on the screen. You may have similar ones in your journal. Discuss: What do you need to know to determine the best path?
	Say: Distance! If we know the distance between each house, we can make a better decision about which path to take. In this example, the first option is shortest, so that's the path we would choose.
	Discuss: What if we had a lot more places to visit? How would we determine the best path? Discussion Goal: The goal here is for students to come to the realization that they would need to explore all possible paths in order to determine which one is the best.
	Say: This is known as the Traveling Salesperson Problem. For each new place to visit, the number of options for possible paths increases factorially.
	Say: A factorial function is written with the letter "n" followed by an explanation point. Here's how n! works: Multiply all whole numbers from the given number down to the number 1. Note: Talk through the examples and the table. The goal here isn't for students to memorize the math, but to understand that with each new house added, the number of potential paths that have to be checked expands very quickly.
	Say: The Traveling Salesperson is an Optimization Problem. It's not a Decision Problem. We know there is a path. Now we must determine which is the shortest or most efficient path.

Slides	Speaker Notes
	Say: The Traveling Salesperson Problem can be solved with an algorithm, which checks each possible option. BUT, it would take massive amounts of computing power to compare every single option, especially as the number of homes to visit (otherwise known as nodes) increases. Therefore, it would take an unreasonable amount of time for the solution to be calculated for most instances of the problem.
	Say: Welcome to heuristics! Heuristics provide a good enough solution to a problem when an actual solution is impractical or impossible.

Do This: Students now navigate to Level 1 on Code Studio where they will use the Traveling Salesperson Widget to find the "best" path to visit all nodes.

- In their journal, students should write down a plan or **heuristic** for choosing a good path.

Discuss: In pairs, students should share their heuristics and make adjustments to their own as needed.



Widget: Traveling Salesperson

Do This: Now students navigate to Level 2 and use the Widget: Traveling Salesperson with Random Nodes. Students test their heuristic on this widget, keeping a log in their journal of the distance for the path their heuristic finds and the best distance the student finds not using the stated heuristic (i.e. clicking around, or "brute force").

- Note: Click reset to try different paths on the same level. Click "New Level" to generate at new random assortment of nodes.



Widget: Traveling Salesperson - Random Nodes

Discuss: *How did you create your heuristic? Did you change your heuristic after testing it out?*

Discussion Goal: Tease out what factors students used to create their heuristics.

Call on several students to share their heuristics. As a class, discuss which heuristic we think is best, and will give us a "good enough" result for most cases.

Discussion Goal: There are times when the Next-Closest heuristic will not provide the best option, but on average it's a good option.

- Students can reflect back on the paper route problem from earlier in the class. This may have been the option they first suggested.

Display: Show the sample heuristic. This may be one that students already came up with. Discuss whether or not this heuristic is "good enough".

Remarks

- The Traveling Salesperson Problem is an optimization problem. We are attempting to find the best path.

It's also unreasonable because there is not an algorithm that can solve the problem in a reasonable amount of time.

We need to use a heuristic to come up with a solution that is "good enough" for most instances of the problem.

Undecidable Problems

Remarks

 We've learned how we address unreasonable problems. Before we finish our study of problems let's learn about one more type, problems that no algorithm will ever be able to solve. This video is a little tricky, but it gives you a good sense for the way this problem shows up.

 **Video:** Show the video on the halting problem, located on the slide

Teaching Tip

Do I Need to Understand the Halting Problem?: Students don't actually need to understand the Halting Problem or the proof in this video. The main ideas are covered in the takeaways and are fairly simple. There are a few problems, most notably "will this code run?" that we've proven there is no algorithm that will ever be able to determine the correct answer in every case. The proof is interesting but if you are short on time you may opt to skip the video.

 **Display:** Review the main takeaways about undecidable problems.

Wrap Up (10 minutes)

 **Discuss:** *Why is a heuristic acceptable when it doesn't always produce the "best" result?*

Answer: A heuristic is used when it's impossible or impractical to use an algorithm to solve a problem. Therefore, we need something that is good enough on average for most instances. This is where a heuristic shines.

Have students think about their answers on their own, then share with a partner, and then finally discuss responses with the entire room.

Remarks

Great! Heuristics are handy in every day life. Think about using mapping software to find the best route to a destination!

Assessment: Check For Understanding

Check For Understanding Question(s) and solutions can be found in each lesson on Code Studio. These questions can be used for an exit ticket.

Question: In which of the following situations is it most appropriate to use a heuristic solution?

Question: In your own words, explain the difference between undecidable problems and unreasonable time algorithms.



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